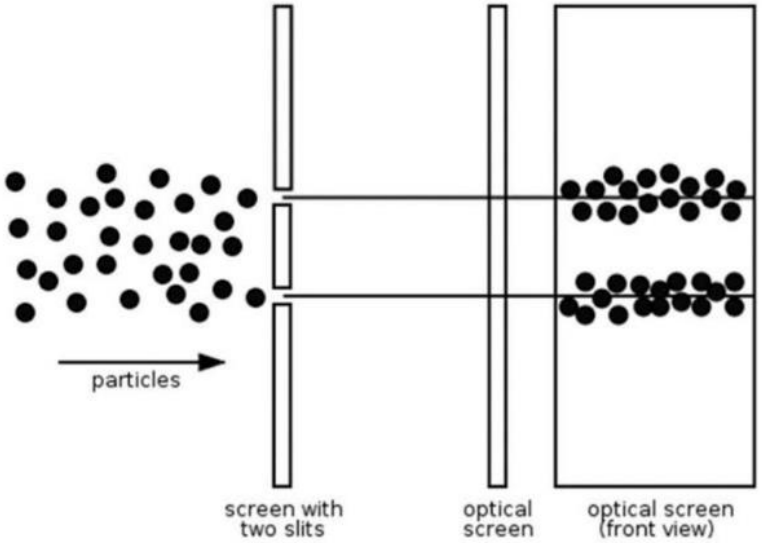
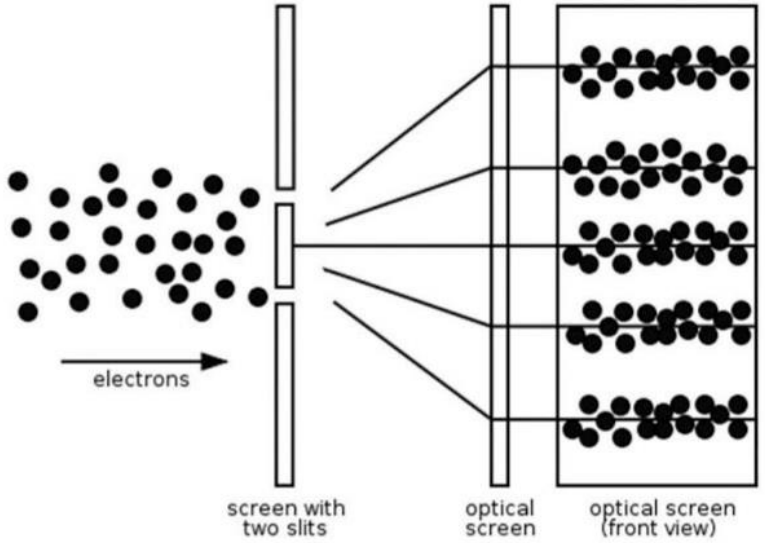
Kvantno procesiranje

Nekonvencionalne platforme in metode procesiranja

Valovna funkcija



detector at slit



no detector at slit

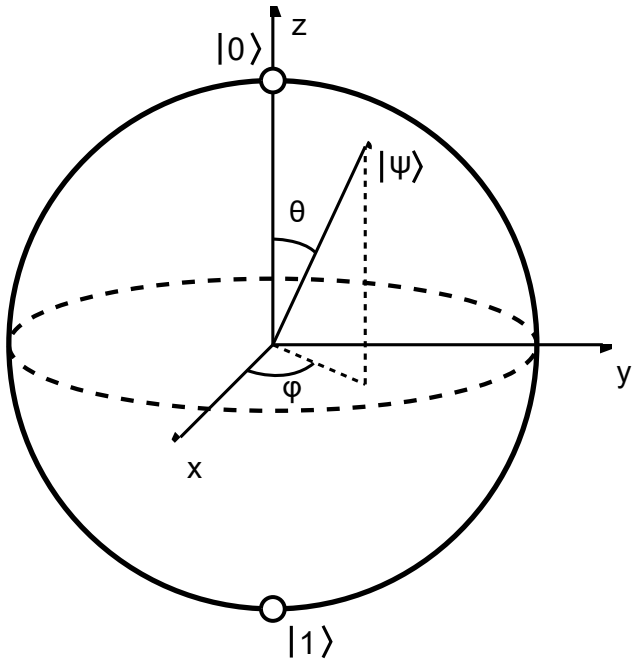
technovedanta.blogspot.com

$$i\hbar \frac{d}{dt}|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle$$

Kvantni bit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\alpha|^2 + |\beta|^2 = 1$$



www.wikipedia.org

$$\alpha = \cos\left(\frac{\theta}{2}\right) \qquad \beta = e^{i\varphi}\sin\left(\frac{\theta}{2}\right)$$

Kvantni register

$$|\boldsymbol{\psi}\rangle = |\psi_{n-1}\rangle \otimes |\psi_{n-2}\rangle \otimes \ldots \otimes |\psi_1\rangle \otimes |\psi_0\rangle$$

$$|\boldsymbol{\psi}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) =$$

$$= \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

Kvantna vrata

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix}\alpha\\ \beta\end{bmatrix}$$

$$|0\rangle = \begin{bmatrix}1\\ 0\end{bmatrix} \qquad |1\rangle = \begin{bmatrix}0\\ 1\end{bmatrix}$$

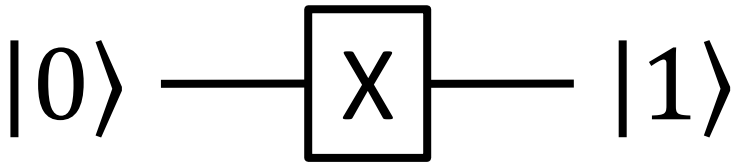
$$I = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}\alpha\\ \beta\end{bmatrix} = \begin{bmatrix}\alpha\\ \beta\end{bmatrix}$$

Negator

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

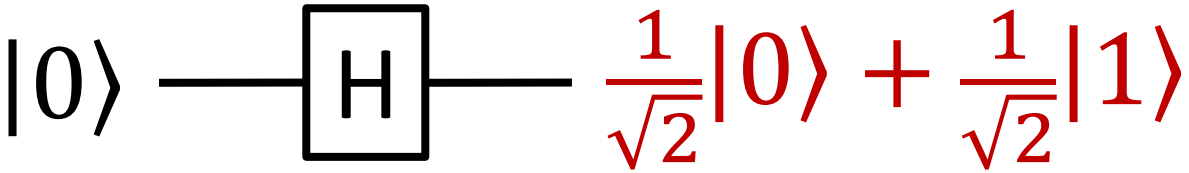
$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}$$



Hadamardova vrata

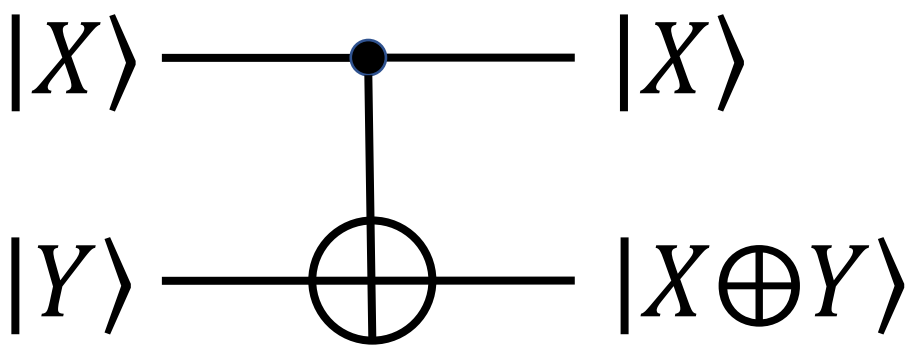
$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

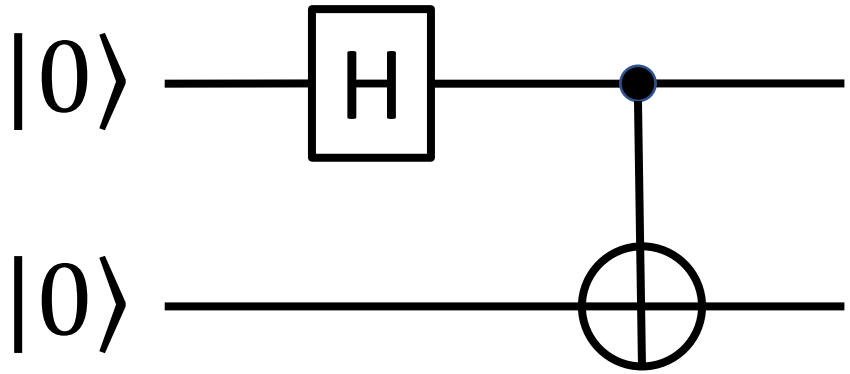


Vrata CNOT

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



Prepleteno stanje



$$\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes |0\rangle =$$

$$= \frac{1}{\sqrt{2}}|00\rangle + 0|01\rangle + \frac{1}{\sqrt{2}}|10\rangle + 0|11\rangle$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

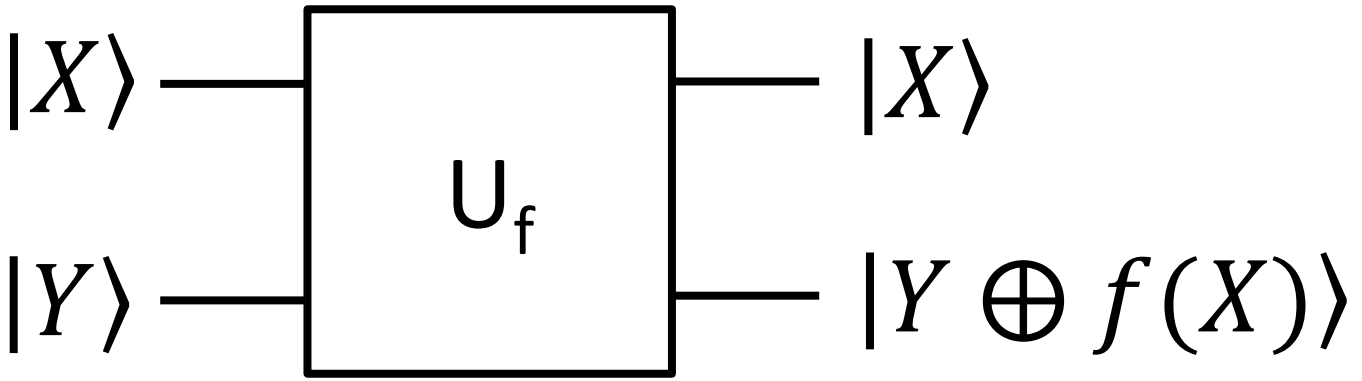
Deutschev algoritem

konstantna funkcija

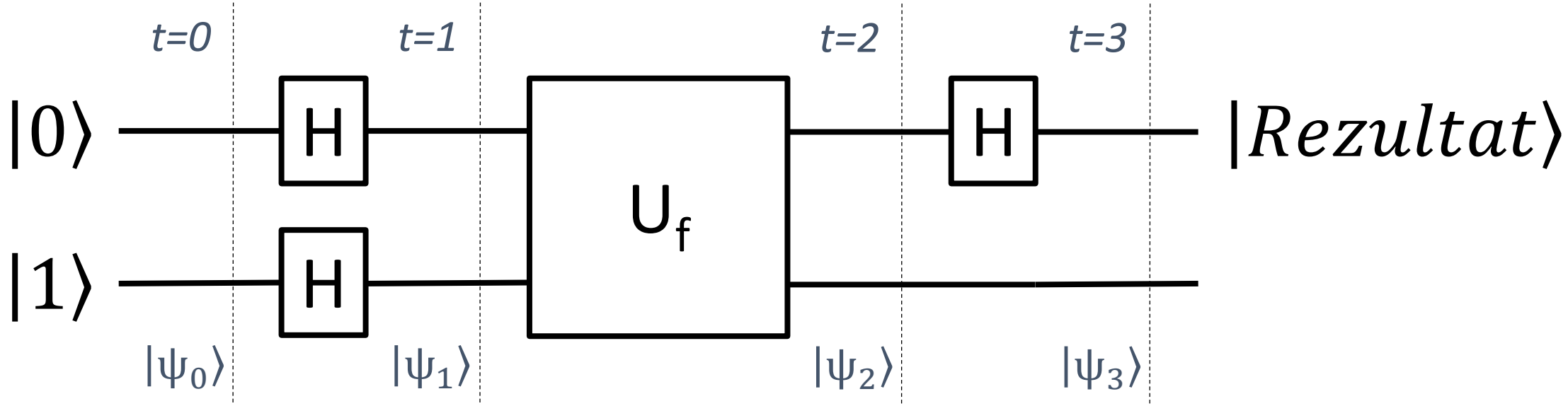
$f(0) = 0$
$f(1) = 0$
ali
$f(0) = 1$
$f(1) = 1$

uravnotežena funkcija

$f(0) = 0$
$f(1) = 1$
ali
$f(0) = 1$
$f(1) = 0$



Implementacija Deutschevega algoritma



$|Rezultat\rangle = |0\rangle \Leftrightarrow f(x)$ je konstantna

$|Rezultat\rangle = |1\rangle \Leftrightarrow f(x)$ je uravnotežena

Analiza Deutschevega algoritma

$$|\psi_0\rangle = |01\rangle$$

$$|\psi_1\rangle = \left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right)$$

$$|\psi_2\rangle = |x\rangle\left(\frac{|0\oplus f(x)\rangle-|1\oplus f(x)\rangle}{\sqrt{2}}\right) = (-1)^{f(x)}\,|x\rangle\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right) =$$
$$= \left(\frac{(-1)^{f(0)}|0\rangle+(-1)^{f(1)}|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right)$$

$$|\psi_2\rangle = \begin{cases} \pm\left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right), & f(x) \text{ je konstantna} \\ \pm\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right), & f(x) \text{ je uravnotežena} \end{cases}$$

$$|\psi_3\rangle = \begin{cases} \pm|0\rangle\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right), & f(x) \text{ je konstantna} \\ \pm|1\rangle\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right), & f(x) \text{ je uravnotežena} \end{cases}$$

Kvantni algoritmi

- algoritem Deutsch-Jozsa
- Simonov algoritem
- Groverjev algoritem
- kvantna Fourierova transformacija
- Shorov algoritem

Uporaba kvantnih računalnikov

- simulacija kvantnih sistemov
- strojno učenje
- kvantno ohlajanje
- kriptografija
- gost prenos informacij
- kvantna teleportacija

Fizične izvedbe

- IBM Q
- Google AI Quantum
- Intel
- Rigetti
- D-Wave

Simulatorji

- seznam
- Quirk
- Quantum Computing Playground

Snovanje

- fizična implementacija
- popravljanje napak
- aplikacije kvantnega procesiranja
- avtomatizacija snovanja
- uporaba qutritov